

REMOTE CONTROL SMOKE CAR
A
Project Report
Submitted In Partial Fulfillment of the Requirements
For The Award Of the
Diploma
In
Electronics and Telecommunication Engineering
Under Guidance Of
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Project Carried Out

At



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- MS Office
- MIT App Inventor
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This is to certify that the dissertation/project proposal entitled "**Remote Controlled Smoke Car**" is done by **Arati Murmu, Aditi Nandi, Anima Samanta, Anupama Chandra, Rana Pal, Rahul Dey, Debraj Goswami, Ganesh Ghosh, Ranit Khan, Soham Sural** an authentic work carried out for the partial fulfillment of the requirements for the award of **Diploma in Electronics and Telecommunication Engineering** under the guidance of **Rahul Sharma**. The matter embodied in this project work has not been submitted earlier forward of any degree to the best of my knowledge and belief.

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This is to certify that this project entitled "**Remote Control Smoke Car**" submitted in partial fulfillment of the degree of **Diploma in Electronics and Telecommunication Engineering** by **West Bengal State Council of Technical Education** through, **K.G Engineering Institute** done by **Arati Murmu, Aditi Nandi, Anima Samanta, Anupama Chandra, Rana Pal, Rahul Dey, Debraj Goswami, Ganesh Ghosh, Ranit Khan, Soham Sural**, Is an authentic work carried out under my guidance & best of our knowledge and belief.

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Certificate of Approval

This is to certify that this documentation of **Industrial Training Program 2023**, entitled "**Remote Control Smoke Car**" is a record of bona-fide work, carried out by **Arati Murmu, Aditi Nandi, Anima Samanta, Anupama Chandra, Rana Pal, Rahul Dey, Debraj Goswami, Ganesh Ghosh, Ranit Khan, Soham Sural** under my supervision and guidance.

In my opinion, the report in its present form is in fulfillment of all the requirements, as specified by the **West Bengal State Council of Technical Education** and as per regulations of the **Ardent Computech Pvt Ltd**. In fact, it has attained the standard, necessary for submission. To the best of my knowledge, the results embodied in this report, are original in nature and worthy of incorporation in the present version of the report for **Diploma in Electronics and Telecommunication Engineering**.

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ACKNOWLEDGEMENT

Success of any project depends largely on the encouragement and guidelines of many others. We take this sincere opportunity to express our gratitude to the people who have been instrumental in the successful completion of this project work.

We would like to show our greatest appreciation to **Rahul Sharma, Technical Trainer at Ardent, Durgapur**. We always feel motivated and encouraged every time by his valuable advice and constant inspiration; without his encouragement and guidance this project would not have materialized. We also want to thank them for sharing their pearls of wisdom with us during the course of this project.

We are also immensely grateful to **Rahul Sir** for his comments on earlier versions of the manuscripts, although any errors are our own and should not tarnish the reputations of these esteemed professionals.

Words are inadequate in offering our thanks to the other trainees, **project assistants** and other members at **Ardent Computech Pvt. Ltd.** for their encouragement and cooperation in carrying out this project work. The guidance and support received from all the members and who are contributing to this project, was vital for the success of this project.

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1. COMPANY PROFILE

ARDENT COMPUTECH PVT. LTD

Ardent Computech Private Limited is an ISO 9001-2015 certified Software Development Company in India. It has been operating independently since 2003. It was recently merged with Ardent Technologies.

Ardent Technologies

Ardent Technologies is a company successfully providing its services currently in UK, USA, Canada and India. The core line of activity at Ardent Technologies is to develop customized application software covering the entire responsibility of performing the initial system study, design, development, implementation and training. It also deals with consultancy services and electronic security systems. Its primary clientele includes educational institutes, entertainment industries, resorts, theme parks, service industry, telecom operators, media and other business houses working in various capacities.

Ardent Collaborations

Ardent Collaborations, the Research Training and Development Department of Ardent Computech Pvt. Ltd. is a professional training Company offering IT enabled services & industrial trainings for B.Tech, MCA, BCA, M.Sc. and MBA fresher's and experienced developers/programmers in various platforms. Summer Training / Winter Training / Industrial training will be provided for the students of B.Tech, M.Tech, MBA, MCA and BCA only. Deserving candidates may be awarded stipends, scholarships and other benefits, depending on their performance and recommendations of the mentors.

Associations

Ardent is an ISO 9001:2015 company. It is affiliated to National Council of Vocational Training (NCVT), Directorate General of Employment & Training (DGET), Ministry of Labour & Employment, and Government of India.

2. ABSTRACT

A remote-controlled vehicle is any mobile machine controlled by means that is physically not connected with origin external to the machine. There are many types in it, based on the controls – radio control device, Wi-Fi controlled and even Bluetooth controlled. These devices are always controlled by humans and take no action autonomously. The main target in such vehicles would be to safely reach a designated point, maneuver the area and reach back to the point of origin. In this project we make use of the Bluetooth technology to control our machine car. We don't call this as a robot as this device doesn't have any sensors. Thereby, sensor less robots are machines. This machine can be controlled by any human using his android mobile phone, by downloading an app and connecting it with the Bluetooth module present inside our car. User can perform actions like moving forward, backward, moving left and right by the means of command using his-her mobile phone app. The task of controlling our car is taken care by the Arduino UNO with micro controller ATMEGA32, 16 MHz processor, 2 KB SRAM (Static Random Accessible Memory) and 32 KB flash memory. Arduino play a major role in the control section and had made it easier to convert digital signals and analogue signals into physical movements. The major reason for using a Bluetooth based tech is that we can change the remote anytime – mobiles phones, tablets and laptops and physical barriers like wall or doors do not affect the car controls.

3. INTRODUCTION

3.1 Background:

Robots are always a fancy topic for students, hobbyists and DIYers. If you are beginner, then building a robot (like a car or an arm) is probably one of the important projects to do after learning about the basics. If you remember the earlier tutorial, I have discussed about HC-05 Bluetooth Module and how to interface one with Arduino. Also, I have provided a simple Bluetooth Controller App, which can be installed on your Android Phone and start transmitting the data. As a continuation to that project, I will be implementing Bluetooth Controlled Robot using Arduino and a few other components and build a simple robotic car that can be controlled using an Android Phone (through an App) over Bluetooth Communication. The robotic car can be controlled wirelessly via a Smartphone. The smartphone has an Android app through which the user can send commands directly to Robot. The robot can move forward, backward, left, and right and can also be stopped. The Arduino's Bluetooth-controlled robot car is interfaced with a Bluetooth module HC-05 or HC-06. We can give specific voice commands to the robot through an Android app installed on the phone. At the receiving side, a Bluetooth transceiver module receives the commands and forwards them to the Arduino, and thus the robotic car is controlled.

3.2 Problem Statement

A remote-controlled vehicle is any mobile machine controlled by means that is physically not connected with origin external to the machine. There are many types in it, based on the controls radio control device, Wi-Fi controlled and even Bluetooth controlled. These devices are always controlled by humans and take no action autonomously. The main target in such vehicles would be to safely reach a designated point, maneuver the area and reach back to the point of origin.

3.3 Motivation

In this project we make use of the Bluetooth technology to control our machine car. We don't call this as a robot as this device doesn't have any sensors. Thereby, sensor less robots are machines. This machine can be controlled by any human using his android mobile phone, by downloading an app and connecting it with the Bluetooth module present inside our car. User can perform actions like moving forward, backward, moving left and right by the means of command using his-her mobile phone app. The task of controlling our car is taken care by the Arduino UNO with micro controller ATMEGA32. Arduino play a major role in the control section and had made it easier to convert digital signals and analogue signals into physical movements. The major reason for using a Bluetooth based tech is that we can change the remote anytime – mobiles phones, tablets and laptops and physical barriers like wall or doors do not affect the car controls

4. OBJECTIVE AND SCOPE

The main goal of our project is to develop a Bluetooth car using wireless technology. In this project Bluetooth module HC-05 is used to connect hardware and software part. With help of that command movement of car is controlled. Thus, we can able to achieve our target.

The project aims are to design an android interface, Arduino bot and write program in to the Arduino microprocessor. Arduino car contains Arduino microcontroller with basic mobility features. Arduino programs contains instructions mediating between android controller and Arduino car.

5. RELATED WORKS

The primary objective is to build an eco-friendly, cost-effective, Bluetooth-controlled robot automobile for material handling. Numerous researchers had built and described robotic devices designed to reduce human labor. These robotic designs were managed by software applications. Winter et al. created a robot controlled by Android. This method was utilized for wireless information transfer [2]. In contrast, the robotic system was constructed utilizing an 8051 microcontroller, Bluetooth module, and a surveillance camera [3]. Using Android and Bluetooth, Guardi et al. developed a communication-based robotic platform [4]. The Cost-Effective Bluetooth-Controlled Robot Car for Material Handling is designed not only to reduce human labor, but also for security and rescue missions [5–7]. Braun et al. [8] described a navigation system for autonomous off-road robots that rely on navigation to avoid obstacles on a map. Arduino was utilized by Tezel and Hangun to design and implement a Bluetooth-controlled robot. Using linear interpolation, sensor data were obtained in this collision-free architecture [9]. Additionally, the pick-and-drop robot was invented. It can be extremely beneficial for defense purposes such as defusing a bomb or land mines [10]. Also invented was the Android based robotic automobile with Bluetooth module [11]. Ritika Pahuja and Narender Kumar built an Arduino based robot automobile operated by Android. When the Android application is activated, the Bluetooth connection is established. Android offers an intuitive interface [12]. Another Android-based automobile with an eco-friendly automation system was designed [13]. Avoiding human mistake, techniques for switching between manual and autonomous driving were described [14]. Shivaprasad B. created a robot that may be used for cleaning fertilizer equipment. Utilizing a sensor on the robot, the designed system gathered vital data [15]. At another design, robotics was incorporated.

6. SYSTEM ANALYSIS

IDENTIFICATION OF NEED

System analysis is a process of gathering and interpreting facts, diagnosing problems and the information to recommend improvements on the system. It is a problem-solving activity that requires intensive communication between the system users and system developers. System analysis or study is an important phase of any system development process. The system is studied to the minutest detail and analyzed. The system analyst plays the role of the interrogator and dwells deep into the working of the present system. The system is viewed as a whole and the input to the system are identified. The outputs from the organization are traced to the various processes. System analysis is concerned with becoming aware of the problem, identifying the relevant and decisional variables, analysis and synthesizing the various factors and determining an optimal or at least a satisfactory solution or program of action.

A detailed study of the process must be made by various techniques like interviews, questionnaires etc. The data collected by these sources must be scrutinized to arrive to a conclusion. The conclusion is an understanding of how the system functions. This system is called the existing system. Now the existing system is subjected to close study and problem areas are identified. The designer now functions as a problem solver and tries to sort out the difficulties that the enterprise faces. The solutions are given as proposals. The proposal is then weighed with the existing system analytically and the best one is selected. The proposal is presented to the user for an endorsement by the user. The proposal is reviewed on user request and suitable changes are made. This is a loop that ends as soon as the user is satisfied with the proposal.

Feasibility Study

Feasibility study is the detailed study expanded from the result of initial investigation. This is done by investigating the existing system in the area under investigation or generally ideas about a new system. A feasibility study of a system proposal is according to its workability, which is the impact on the organization, ability to meet their user needs and effective use of resources.

Thus, when a new application is proposed it normally goes through a feasibility study before it is approved for development. Objective of feasibility study is to acquire a sense of the scope of the problem. It is carried out to select the best system that meets performance requirements.

Feasibility analysis involves the following steps:

From a project and appoint a project leader.

- Prepare system flowchart.
- Weigh system performance.
- Prepare and report final project directive to management.

The document provides the feasibility of the project that is being designed and lists various area that were considered very carefully during the feasibility study of this project such as Technical, Economic and operational feasibilities.

Economic Feasibility

Are there sufficient benefits in creating the system to make the costs acceptable? Or, are the costs of not creating the system so great that it is advisable to undertake the project? These are the important questions to be answered in economic feasibility. Proposed system will not face any economic constraint as it will be developed by students only. And it will help to automate manual processing, which will save time and money. This will provide economic benefits.

Operational Feasibility

Will the system be used if it is developed and implemented? Will there be resistance from users that will undermine the possible application benefits? When we look from this perspective, we don't see any risk in implementing and making it operational. This project will help everyone, as it is planned to be more reliable, maintainable, affordable and producible.

Technical Feasibility

Can the work for the project be done with current equipment, existing software and hardware technology and available personnel? These are the questions which need to be answered to check Technical feasibility. Project will be developed with Arduino which is the microcontroller. There is basic requirement of hardware to run this application. This application can be accessed by using any device like (Personal Computers, Laptop and with some hand-held devices).

7. WORK FLOW

The system design is broken down into two main categories,

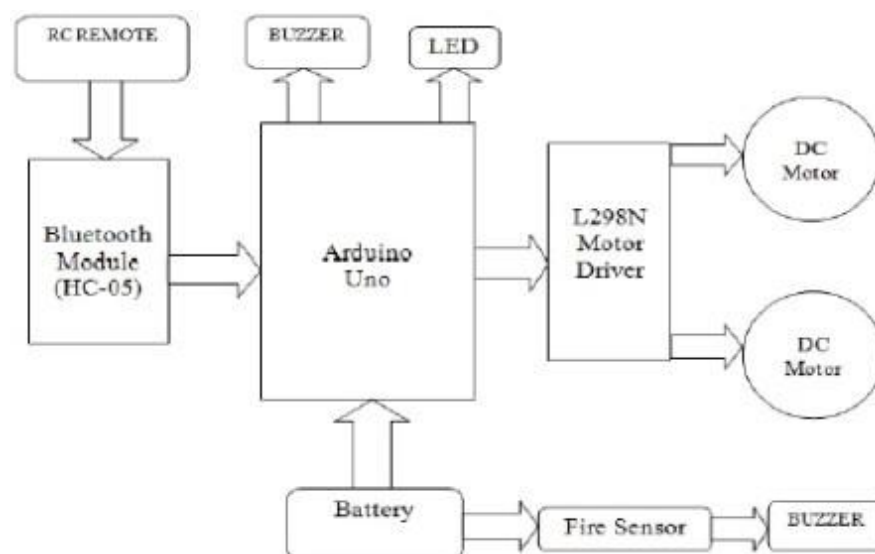
Power Supply: A power supply is an electronic device that supplies electric energy to an electrical load. The primary function of a power supply is to convert one form of electrical energy to another and, as a result, power supplies are sometimes referred to as electric power converters.

Bluetooth module: It is small wireless serial communication module that can be connected with a Micro-Controller to receive and send data when connected with other Bluetooth devices.

Arduino-UNO: Arduino is an open-source prototyping platform based on easy-to-use hardware and software. Arduino consists of both a physical programmable circuit board and a piece of software, or IDE (Integrated Development Environment) that runs on your computer, used to write and upload computer code to the physical board.

Motor driver: It is a small circuit that hoists the motor driving IC, and can control two motors at the same time. It controls the motor speed by pulse width modulation (PMW).

Gas or Fire Sensor: Fire or Gas sensors arranged to monitor and annunciate the status of **fire** or presence of **gas alarm** and to initiate appropriate response to those signals.



8. HARDWARE & SOFTWARE REQUIREMENTS

Software Requirements

- **OS:** Windows 10 or above
- **MIT App Inventor**
- **Arduino IDE:** Arduino 1.8.9 or above

Computer Hardware Components

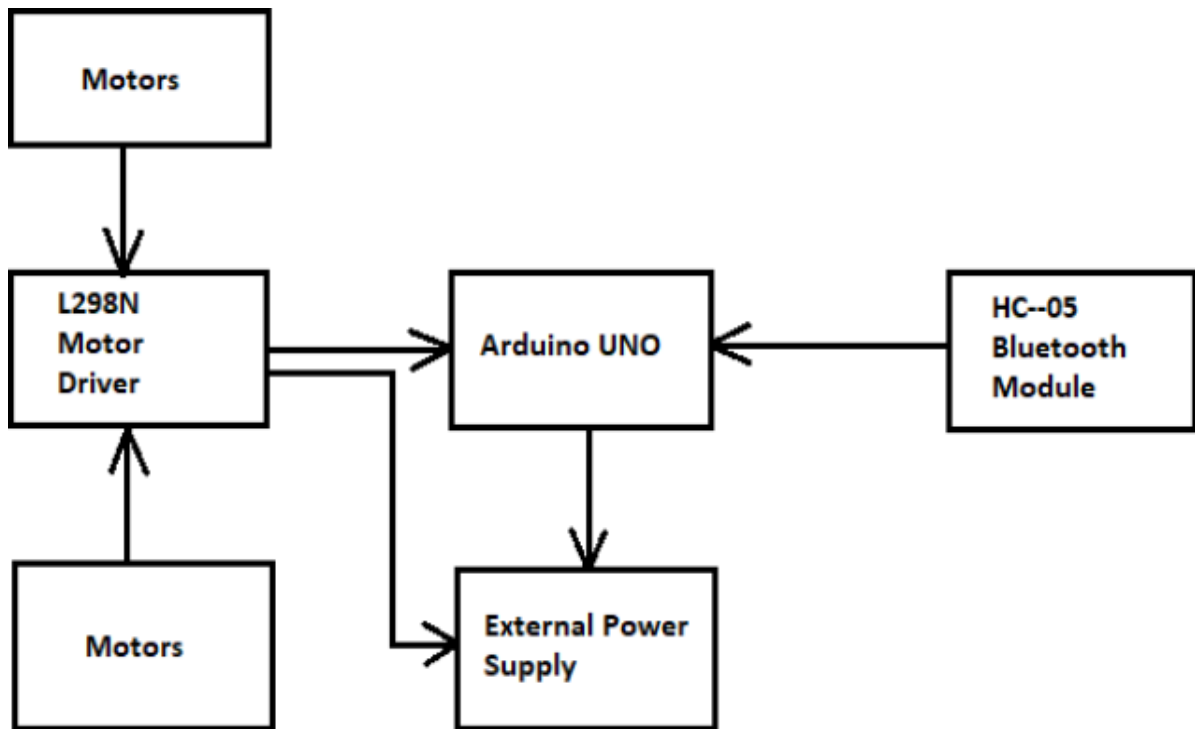
- **Processor:** i5 7th Generation.
- **Hard Disk:** 1 TB
- **Memory:** 8GB RAM
- **Mouse:** Any Standard.
- **Keyboard:** Any Standard.
- **Monitor:** Any color monitor.

Robot Hardware Parts

- Arduino Board and Cable
- **Wires:** Jumper Wires as required.
- **Sensors:** HC 05, MQ 135
- **Others:** L298N Motor Driver, 9V DC Motor

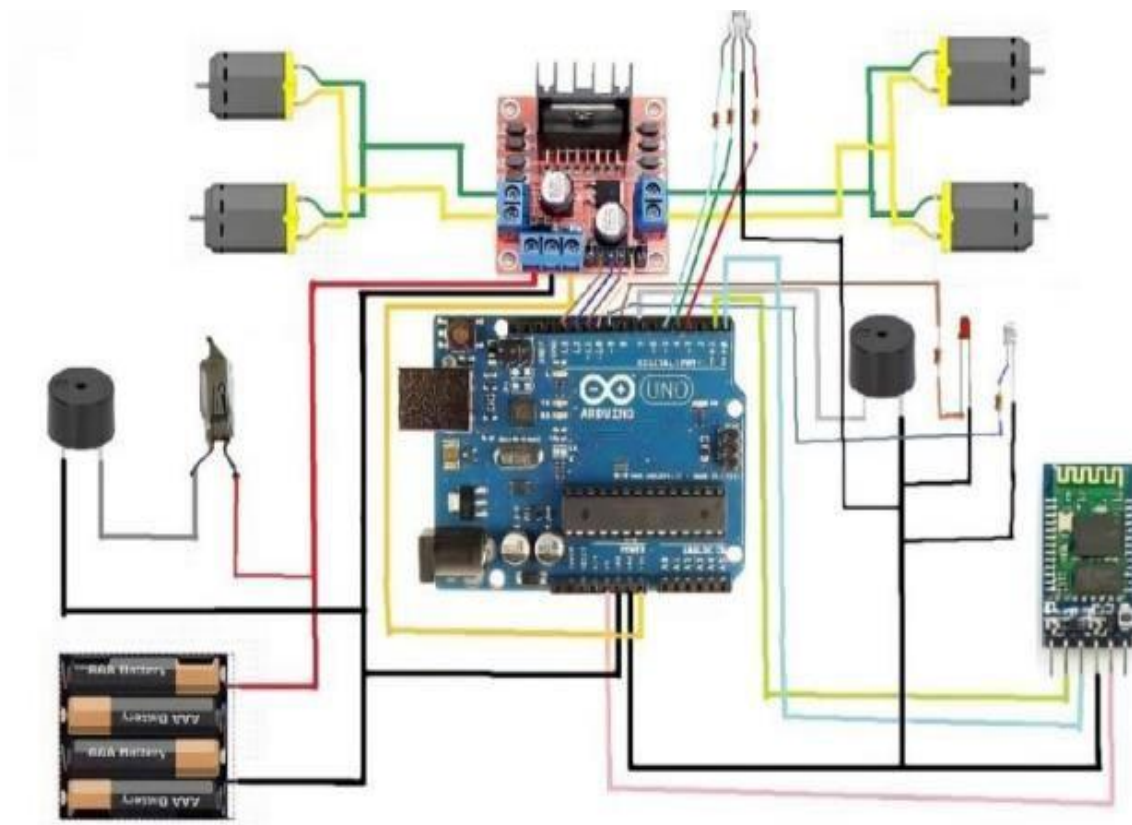
9. SYSTEM ARCHITECTURE

9.1 Block Diagram



Block Diagram

9.2 Circuit Diagram and Description



As shown in the diagram, two rechargeable batteries as supply is used which is connected to motor driver and Arduino respectively. When the circuit is energized, we will have to first pair the Android phone with the Bluetooth module through the phone's Bluetooth settings. The default password of the Bluetooth module will be '1234'. Once the phone gets paired, open the application 'CAR BLUETOOTH RC' which we can download from 'Google play store'. On opening the application, there will be a set of controls displayed on the screen. If the device has not been connected, the control will be locked, meaning the control buttons cannot be pressed. When the car is at its initial position, the application automatically sends the command 'S' meaning stop. The stop command is put in a loop that keeps on repeating throughout the execution of the program. As the user presses any control buttons, the stop command will be interrupted by the move forward, backward, right, left, depending on the user, and the car moves

like wise. The program is designed in such a manner that we can also give two commands at the same time i.e. move front and turn right or left and same with the backward motion. The Arduino also stores the program in its memory so it does not require re-uploading of Program. The IN1, IN2, IN3 and IN4 are the inputs for the motor driver that receives command from the Arduino for the two motors respectively. The motor driver should be grounded with the Arduino ground pin (GND). The motor driver requires minimum of 6v and above to run, any voltage below 6v the motor remains off. The RXD pin of the Bluetooth module is for receiving commands from the Android devices and sends to Arduino through this pin and the TXD is for transmitting or sending dates or information's. It is supplied with a 5v dc source from the arduino 5v pin. The main part of the above circuit diagram is arduino UNO. The power supply section is very Important. It should provide constant voltage to the devices for successful working of the project.

Arduino Uno

Arduino.cc designed the Arduino Uno based on the Microchip ATmega328P microprocessor. The board has digital and analogue I/O pins for interfacing with shields and other circuits. The Arduino board includes 14 digital I/O pins (six may output PWM), 6 analogue Input Output (I/O) pins, and is programmable through a type B USB connector using the Arduino IDE. This board has a USB interface; thus, a USB connection connects it to a computer and Arduino IDE software is used to programmed it.



Hc-05 Bluetooth Module

As it pertains to technology and communication, wireless communication is rapidly replacing conventional connections. Designed to replace cable connections, HC-05 communicates with devices via serial transmission. Typically, it is used to exchange files between small devices such as mobile phones via a short-range wireless connection. It operates inside the 2.45GHz frequency band. The data transfer rate is variable up to 1Mbps and has a range of 10 meters. The HC-05 module may work within a voltage range of 4-6V. It supports 9600, 19200, 38400, 57600, etc. baud rates. Importantly, it can be operated in Master Slave mode, which prevents it from transmitting or receiving data from other sources. If you've ever wanted to operate your Mechanical Bots with an Android Phone or create the robots with a bespoke remote, this guide will teach you how to use a Bluetooth Module HC-05 for these and other purposes



L298n Motor Driver

L298N is a dual full-bridge motor driver module for DC and stepper motors. It controls two DC motors' speed and direction. This module contains a dual-channel H-Bridge motor driver IC. This module controls DC motor speed and direction using two methods. PWM controls speed, and H-Bridge controls rotation. These modules may control two DC motors or one stepper. This module has an L298 motor driver IC and 78M05 5V regulator. L298N Module controls up to 4 DC motors or 2 with directional and speed control.



Buzzer

A buzzer is a small but effective sound component. Its modest 2-pin structure makes it easy to utilize on breadboard, Perf Board, and PCBs, making it a popular electrical component. This buzzer is powered by a 4V to 9V DC source. A 9V battery can be utilized, however a +5V or +6V DC source is preferable. The buzzer is usually connected to a switching circuit that turns it ON and OFF at set times. Positive Symbol indicates longer terminal lead. For 6V DC-capable. Negative terminal lead is Usually grounded.



Gas Sensor

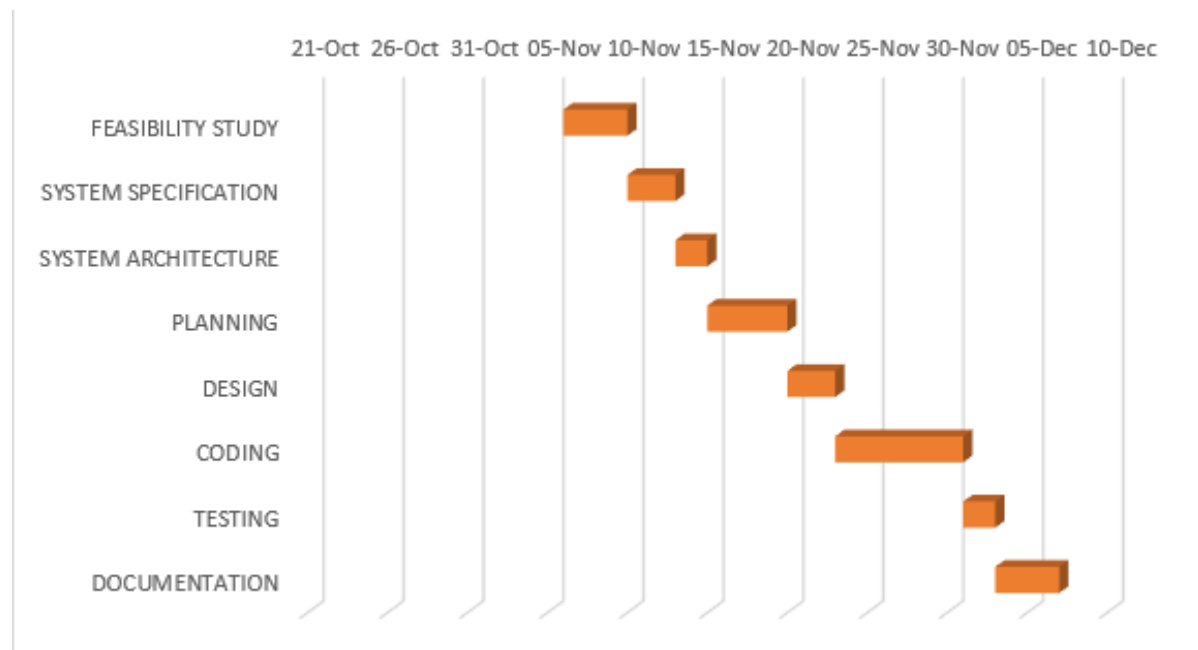
The **MQ-135 Gas sensor** can detect gases like Ammonia (NH_3), sulfur (S), Benzene (C_6H_6), CO_2 , and other harmful gases and smoke. Similar to other MQ series gas sensor, this sensor also has a digital and analog output pin. When the level of these gases go beyond a threshold limit in the air the digital pin goes high. This threshold value can be set by using the on-board potentiometer. The analog output pin, outputs an analog voltage which can be used to approximate the level of these gases in the atmosphere.



10. SYSTEM DESIGN

10.1 Gantt Chart

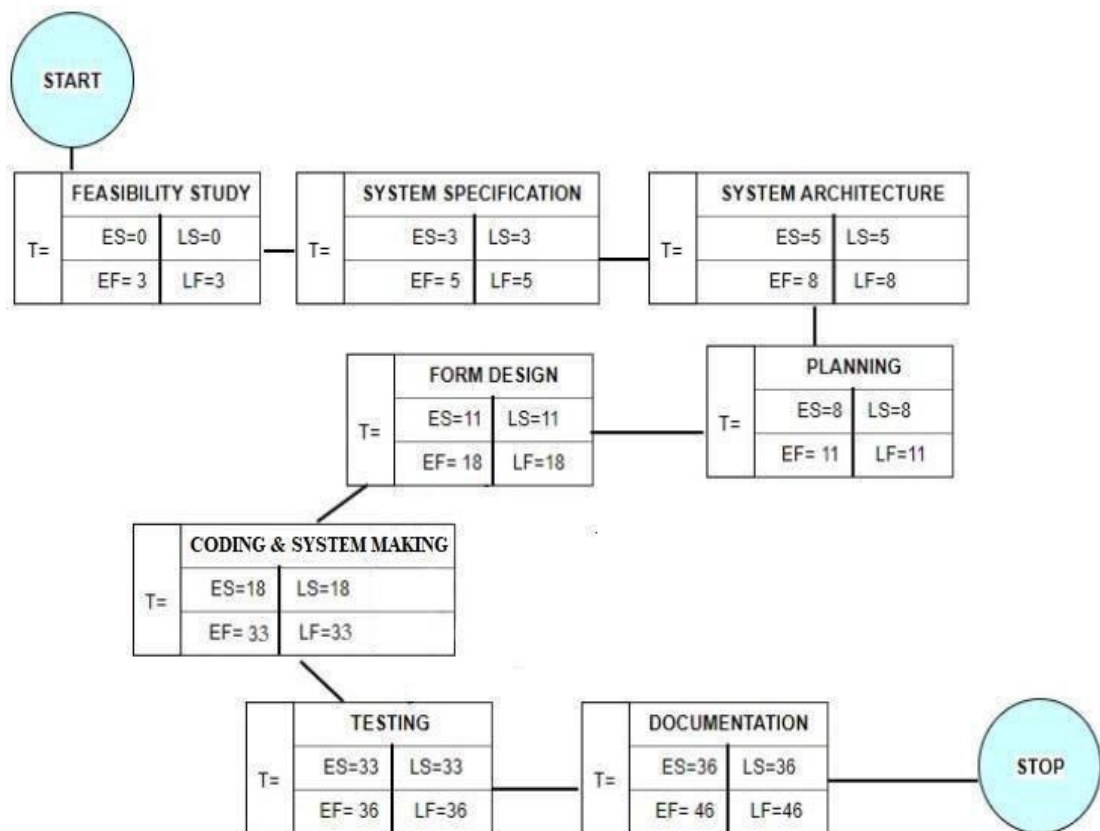
TASK	START DATE	DURATION
Feasibility Study	11-Dec	4
System Specification	15-Dec	3
System Architecture	17-Dec	2
Planning	19-Dec	2
Design	22-Dec	3
Coding	24-Dec	2
Testing	26-Dec	2
Documentation	28-Dec	2



10.2 Pert chart

TASK	START DATE	DURATION
Feasibility Study	11-Dec	4
System Specification	15-Dec	3
System Architecture	17-Dec	2
Planning	19-Dec	2
Design	22-Dec	3
Coding	24-Dec	2
Testing	26-Dec	2
Documentation	28-Dec	2

Pert chart Data



Pert Chart

10.3 Sequence Diagram

Sequence diagrams can be useful reference diagrams for businesses and other organizations. Try drawing a sequence diagram to:

- Represent the details of a UML use case.
- Model the logic of a sophisticated procedure, function, or operation.
- See how tasks are moved between objects or components of a process.
- Plan and understand the detailed functionality of an existing or future scenario.

Popular Sequence Diagram Uses

- **Usage Scenario** – A usage scenario is a diagram of how your system could potentially be used. It's a great way to make sure that you have worked through the logic of every usage scenario for the system.
- **Method Logic** - Just as you might use a UML sequence diagram to explore the logic of a use case, you can use it to Usage Scenario - A usage scenario is a diagram of how your system could potentially be used. It's a great way to explore the logic of any function, procedure, or complex process.

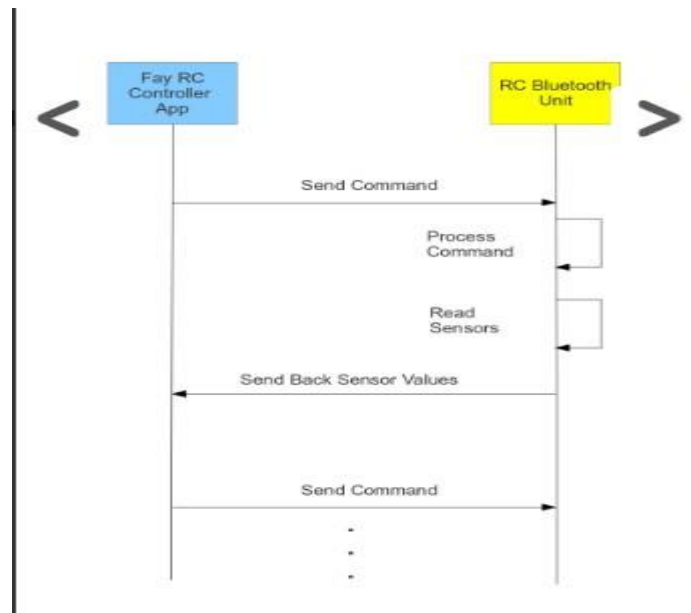


Figure: Sequence Diagram

10.4 Activity Diagram

Activity diagram is another important diagram in UML to describe the dynamic aspects of the system. Activity diagram is basically a flowchart to represent the flow from one activity to another activity. The activity can be described as an operation of the system. The control flow is drawn from one operation to another. This flow can be sequential, branched, or concurrent.

Activity diagrams deal with all type of flow control by using different elements such as fork, join, etc.

Purpose of Activity Diagrams

The basic purpose of activity diagrams is similar to other four diagrams. It captures the dynamic behavior of the system. Other four diagrams are used to show the message flow from one object to another but activity diagram is used to show message flow from one activity to another.

Where to Use Activity Diagrams?

The basic usage of activity diagram is similar to other four UML diagrams. The specific usage is to model the control flow from one activity to another. This control flow does not includes messages. Activity diagram is suitable for modeling the activity flow of the system. An application can has multiple systems. Activity diagram also captures these systems and describes the flow from one system to another. This specific usage is not available in other diagrams. These systems can be database, external queues, or any other system.

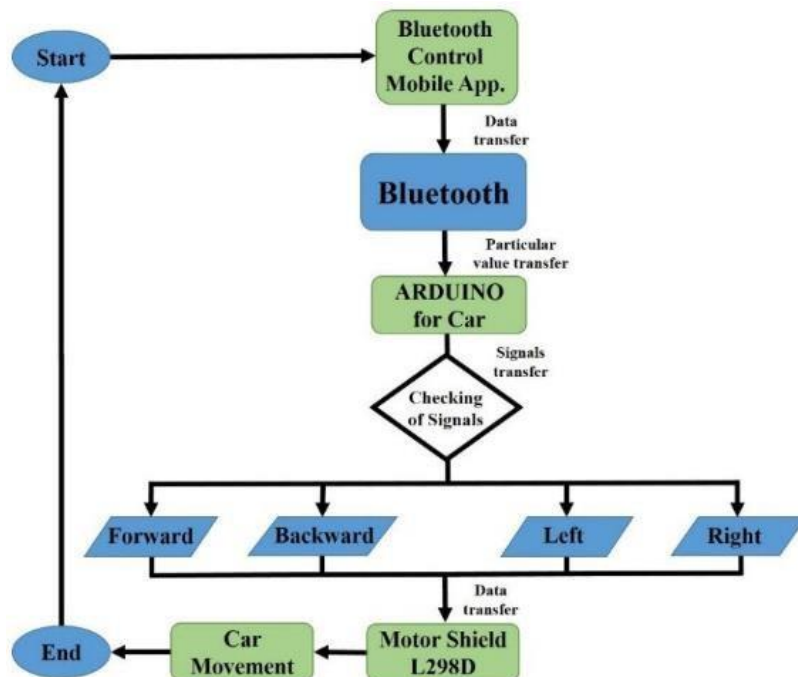
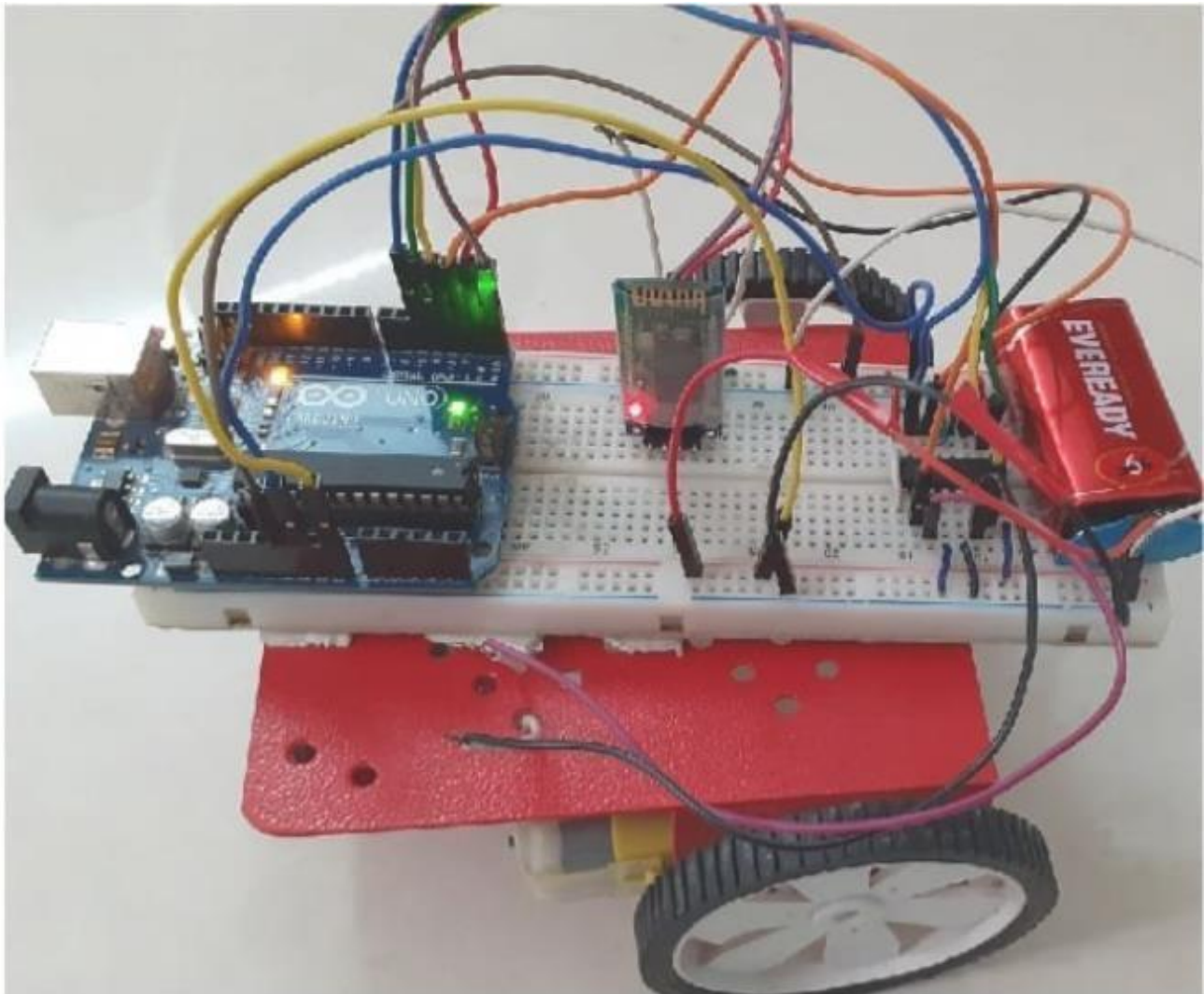
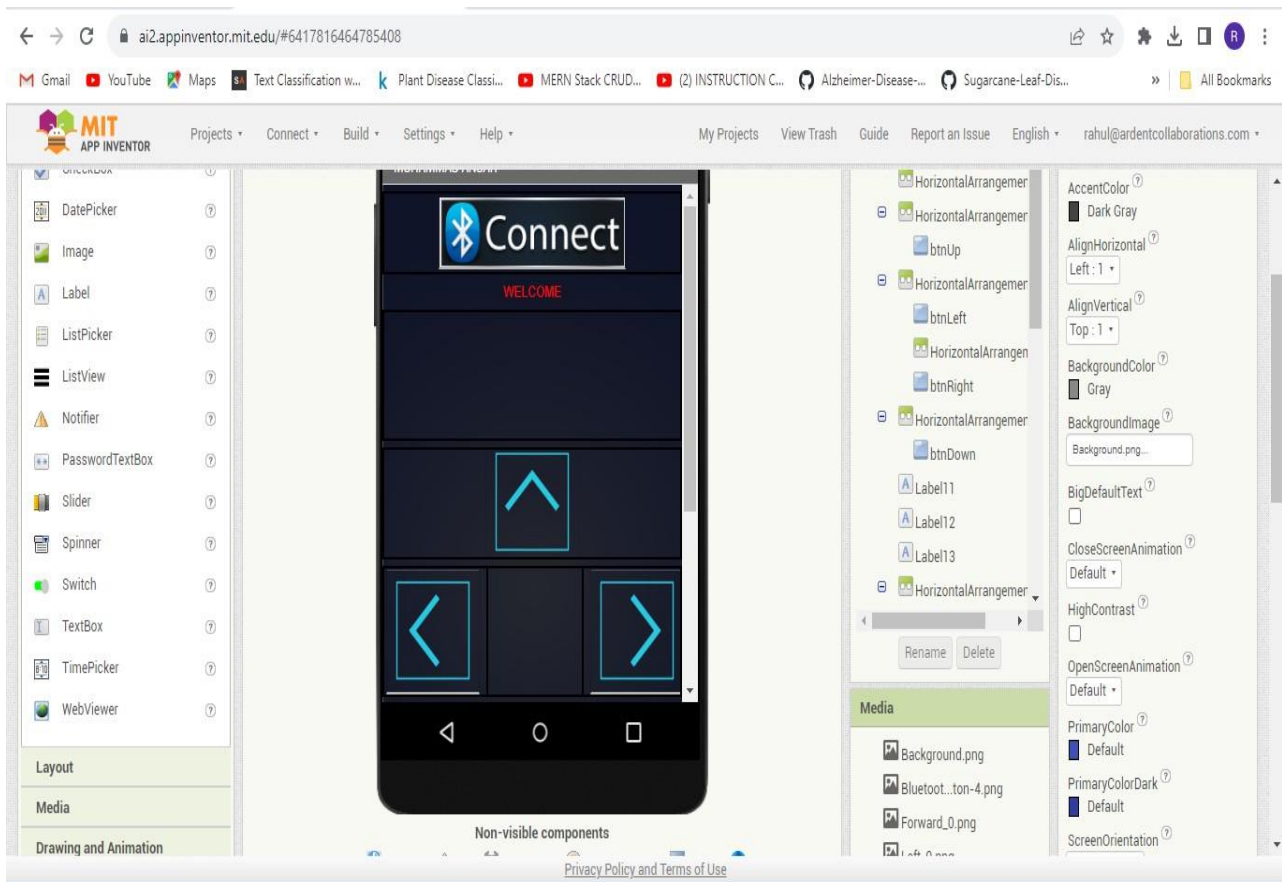


Figure: Activity Diagram

10.5 Snapshots





11. IMPLEMENTATION & TESTING

A software system test plan is a document that describes the objectives, scope, approach and focus of software testing effort. The process of preparing a test plan is a usual way to think the efforts needed to validate the acceptability of a software product. The complete document will help people outside the test group understand the "WHY" and "HOW" of product validation. It should be thorough enough to be useful but not so thorough that no one outside the test group will read it.

INTRODUCTION

Testing is the process of running a system with the intention of finding errors. Testing enhances the integrity of a system by detecting deviations in design and errors in the system. Testing aims at detecting error-prone areas. This helps in the prevention of errors in a system. Testing also adds value to the product by conforming to the user requirements. The main purpose of testing is to detect errors and error prone areas in a system. Testing must be through well planned. A partially tested system is to detect errors and error prone areas in a system. Testing must be through well planned. A partially tested system is as bad as an untested system. And the price of an untested and under tested system is high.

OBJECTIVES OF TESTING

The objective our test plan is to find and report as many bugs as possible to improve the integrity of our program. Although exhaustive testing is not possible, we will exercise a broad range of tests to achieve our goal. Our user interface to utilize these functions is designed to be user- friendly and provide easy manipulation of the tree. The application will only be used as a demonstration tool, but we would like to ensure that it could be run from a variety of platforms with little impact on performance or usability.

PROCESS OVERVIEW

The following represents the overall flow of the testing process:

- Identify the requirements to be tested. All test cases shall be derived using the current program specifications.
- Identify which particular test(s) will be used to test each module.
- Review the test data and test cases to ensure that the unit has been thoroughly verified and that the test data and test cases are adequate to verify proper operation of the unit.

- Identify the expected results for each test.
- Document the test case configuration, test data, and expected results.
- Document the test data, test cases, and test configuration used during the testing process. This information shall be submitted via the Unit/System Test Report (STR).
- Successful unit testing is required before the unit is eligible for component integration/system testing.
- Unsuccessful testing requires a Bug Report Form to be generated. This document shall describe the test case, the problem encountered, possible cause, and the sequence of events that led to the problem. It shall be used as a basis for later technical analysis.

TEST CASES

A test case is a document that describes an input, action, or event and expected response, to determine if a feature of an application is working correctly. A test case should contain particular such as test case identifier, test condition, input data. Requirement expected results. The process of developing test cases can help find problems in the requirements or design of an application since it requires completely thinking through the operations of the application.

TESTING STEPS

- **Unit Testing:** Unit testing focuses efforts on the smallest unit of software design. This is known as module testing. The modules are tested separately. The test is carried out during programming stage itself. In this step, each module is found to be working satisfactory as regards to the expected output from the module.
- **Integration Testing:** Data can be lost across an interface. One module can have an adverse effect on another, sub functions, when combined, may not be linked in desired manner in major functions. Integration testing is a systematic approach for constructing the program structure, while at the same time conducting test to uncover errors associated within the interface.

VALIDATION

At the culmination of the integration testing, Software is completely assembled as a package. Interfacing errors have been uncovered and corrected and a final series of software test begin in validation testing. Validation testing can be defined in many ways, but a simple definition is that the validation succeeds when the software functions in a manner that is expected by the customer. After validation test has been conducted, one of the three possible conditions exist.

- a) The function or performance characteristics confirm to specification and are accepted.
- b) A deviation from specification is uncovered and a deficiency lists is created.
- c) Proposed system under consideration has been tested by using validation test and found to be working satisfactorily.

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Test Type	Unit Testing
Test Case Number	1
Test Case Name	Training & Testing
Item(s) to be tested	
1	• The user should control the car via Bluetooth and can steer and go forwards and backwards • Smartphone gives a Bluetooth signal to the car and from the signal, the car works. • F is used for forwarding car movement. B is used for a backward car moment. Similarly, Other characters are used for other directions which are mentioned in the Arduino code
Specifications	
Input • Signals from Phone • When car is placed in smoke zone	Expected Output/Result • Car moves according to the desired signal • Buzzer produces sound as smoke is detected by sensor

WHITE BOX TESTING

In white box testing, the UI is bypassed. Inputs and outputs are tested directly at the code level and the results are compared against specifications. This form of testing ignores the function of the program under test and will focus only on its code and the structure of that code. Test case designers shall generate cases that not only cause each condition to take on all possible values at least once, but that cause each such condition to be executed at least once. To ensure this happens, we will be applying Branch Testing. Because the functionality of the program is relatively simple, this method will be feasible to apply.

BLACK BOX TESTING

Black box testing typically involves running through every possible input to verify that it results in the right outputs using the software as an end-user would. We have decided to perform Equivalence Partitioning and Boundary Value Analysis testing on our application.

SYSTEM TESTING

The goals of system testing are to detect faults that can only be exposed by testing the entire integrated system or some major part of it. Generally, system testing is mainly concerned with areas such as performance, security, validation, load/stress, and configuration sensitivity.

OUTPUT TESTING

After performing the validation testing, the next step is output testing of the proposed system, since no system could be useful if it does not produce the required output in a specific format. The output format on the screen is found to be correct. The format was designed in the system design time according to the user needs. For the hardcopy also; the output comes as per the specified requirements by the user. Hence output testing did not result in any correction for the system.

USER ACCEPTANCE TESTING

User acceptance of a system is the key factor for the success of any system. The system under consideration is tested for the user acceptance by constantly keeping in touch with the prospective system users at the time of developing and making changes whenever required.

This is done in regard to the following point:

- a) Input Screen Design.
- b) Output Screen Design.
- c) Format of reports and other outputs.

INTEGRATION TESTING

Software testing is always used in association with verification and validation. In the testing phase of this project our aim is to find the answer to following two questions.

- Whether the software matches with the specification (i.e. process base) to verify the product.
- Whether this software in one client what wants (i.e. product base) to validate the product.
- Unit testing and integration testing has been carried out to find the answer to above questions. In unit testing each individual module was test to find any unexpected behavior if exists. Later all the module was integrated and flat file was generated.

FUNCTIONAL TESTING

These are the points concerned during the stress test:

- Nominal input: character is in putted in the place of digits and the system has to flash the message "Data error".
- Boundary value analysis: exhaustive test cases have designed to create an output report that produces the maximum (and minimum) allowable number of table entries.

12. SYSTEM SECURITY MEASURES

DATABASE SECURITY

System security measure is meant to be provided to make your system reliable and secured from unauthorized user may create threats to the system. So, you should follow some security measures. We have used security levels in database level at system level.

SYSTEM SECURITY

Robotics can involve adding internet connectivity to a system of interrelated computing devices, mechanical and digital machines, objects, animals and/or people. Each "thing" is provided a unique identifier and the ability to automatically transfer data over a network. Allowing devices to connect to the internet opens them up to a number of serious vulnerabilities if they are not properly protected.

LIMITATIONS

The L298N is specified to work with 7-12V, so we are using a bigger battery with 7.2V or 11.1V. When the new Power source is connected, everything turns on and you can connect your phone with the Bluetooth device, but if the power source is below or above it then this might cause the motor driver to burn.

7. COST ESTIMATION

Boehm proposed COCOMO (Constructive Cost Estimation Model) in 1981. COCOMO is one of the most generally used software estimation models in the world. COCOMO predicts the efforts and schedule of a software product based on the size of the software

The necessary steps in this model are:

1. Get an initial estimate of the development effort from evaluation of thousands of delivered lines of source code (KDLOC).
2. Determine a set of 15 multiplying factors from various attributes of the project.
3. Calculate the effort estimate by multiplying the initial estimate with all the multiplying factors i.e., multiply the values in step 1 and step 2.

The initial estimate (also called nominal estimate) is determined by an equation of the form used in the static single variable models, using KDLOC as the measure of the size. To determine the initial effort E_i in person-months the equation used is of the type is shown below

$$E_i = a * (KDLOC)^b$$

The value of the constant a and b are depending on the project type. In COCOMO, projects are categorized into three types:

1. Organic
2. **Semidetached**
3. Embedded

1. Organic: A development project can be treated of the organic type, if the project deals with developing a well-understood application program, the size of the development team is reasonably small, and the team members are experienced in developing similar methods of projects. Examples of this type of projects are simple business systems, simple e-learning portal, and data processing systems.

2. Semidetached: A development project can be treated with semidetached type if the development consists of a mixture of experienced and inexperienced staff. Team members may have finite experience in related systems but may be unfamiliar with some aspects of the order being developed. Example of Semidetached system includes developing a new operating system (OS), a Database Management System (DBMS), and complex e-learning portal.

3. Embedded: A development project is treated to be of an embedded type, if the software being developed is strongly coupled to complex hardware, or if the stringent regulations on the operational method exist. For Example: ATM, Air Traffic control.

For three product categories, Boehm provides a different set of expression to predict effort (in a unit of person month) and development time from the size of estimation in KLOC (Kilo Line of code) efforts estimation takes into account the productivity loss due to holidays, weekly off, coffee breaks, etc.

According to Boehm, software cost estimation should be done through three stages:

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1. Basic Model
2. *Intermediate Model*
3. Detailed Model

1. Basic COCOMO Model: The basic COCOMO model provides an accurate size of the project

parameters. The following expressions give the basic COCOMO estimation model:

Effort applied to the project: $E = a_b(KLOC)^{b_b}$ (in Person-month)

Development time: $D = c_b(E)^{d_b}$ (in month)

Manpower required: $P = \frac{E}{D}$ (in Person)

Where a_b, b_b, c_b, d_b are constants for each category of software product

Software Project	A	B	C	D
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3.0	1.12	2.5	0.35
Embedded	3.6	1.20	2.5	0.32

Software project using semi-detached mode with 4 Kloc.

Effort (E) = $a_b(KLOC)^{b_b} = 3.0 \times (0.04)^{1.12} = 0.779\text{PM}$

Development Time (D) = $c_b(E)^{d_b} = 2.5(0.779)^{0.35} = 2.291 \text{ Months(M)}$

Person Required (P) = $E/D = 0.779/0.35 = 0.54 \text{ Persons} \sim 1 \text{ Person}$

Note: With extra manpower the project is completed within 1 month

11. CONCLUSION

Robots and smart phone are a perfect match for us to realize smart living not only at work, but in our homes, with the aid of easily available and widely used technology, the Bluetooth as the mobile devices are becoming more advanced, using them for controlling Robots and other wireless devices is likely to be a huge trend. It can be concluded that this idea of smart living will let us control our surroundings remotely and wirelessly. With the ever-increasing problems, our knowledge has to expand to adapt better to the changes all around us. In the same way it is hoped that this activity is a small step that would lead us to further enhancements and goals.

12. FUTURE SCOPE AND FURTHER ENHANCEMENTS

In future we can interface sensors to this robot so that it can monitor some other parameters and can improve the efficiency using Internet of Things (IoT) technology. We can also add wireless cameras, in order to incorporate other security features. It can also realize smart living, more specifically the home lighting control system using Bluetooth Technology. The robot can also be used for surveillance. The surveillance has always been a quite sensitive task that includes so many risks. So, it's better to use robots for this job instead of people. This work can be further modified for material handling in industries also.

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